PHYS 6751, Graduate Laboratory: Nuclear and Particle Physics: Fall 2024

Syllabus, August 26, 2024

Instructor:	Dr. Paul M. King	Call Number:	11294
Office:	207 Edwards	Credit Hours:	3 credits
Phone:	(740) 593 1974	Class Times:	Mon. & Wed., 2:00 - 3:50PM
Email:	kingp1@ohio.edu	Class Location:	Edwards, rooms 208 and 101
Office Hours:	By appointment		

This will be a course on laboratory techniques, thus most of our time will be spent in the laboratory. However, a portion of our time, mostly at the beginning of the term, will be spent in a seminar to discuss general aspects of nuclear physics experiments, instrumentation, and analysis. Active participation in both laboratory and seminar components is expected from all students. The course expectations as described in this syllabus are subject to change as needed to account for unforeseen circumstances. In that event, a revised syllabus will be distributed and posted on Canvas.

I care about your success in this class. Please let me know if something in the syllabus, or in the course as it develops, makes it difficult for you to learn. We will work together to find a solution.

As part of Ohio University's learning management system transition, this course will be delivered in Canvas. Course materials, discussions, activities, assessments, and assignments, as well as student grades, can be accessed through `https://canvas.ohio.edu`.

1 Learning outcomes

This course will give you experience with instrumentation common in nuclear physics. You should develop basic laboratory skills, such as record-keeping, error analysis, and linear regression, and you should be able to write reports which follow the general format used by professional physicists. At the conclusion of the course, successful students will be able to:

* Construct arguments and identify trends based on experimentally controlled observations;
* Discuss the instrumentation used in an experimental investigation, including any systematic errors or biases that might be introduced by these;
* Work together in small groups to design and construct an experiment;
* Maintain laboratory notebooks of sufficient quality for beginning graduate-level research;
* Use a computer to do sophisticated data analysis, including uncertainty analysis using professional standards;
* Use theoretical models to predict and understand measurement results;
* Make written scientific arguments using a number of standard elements of technical communication;

* Present orally the data, analysis, and conclusions of an experiment, or a literature review of a related topic, in the style of conference presentations in nuclear and particle physics.

2 Course Structure & Assignments

Students will be expected to complete a total of five projects (one analysis tutorial and four experiments), over the course, working in pairs. Each experiment is expected to require about two weeks for proper data collection and analysis.

You should attend all laboratory sessions for the scheduled time. If you have finished your data collection, you should use the time to work on your analysis in Edwards 208. If you would need to leave early, please discuss it with me before leaving.

2.1 Group roles

For this class, you will be working in pairs. In alternating lab sessions, you will be the Recorder or the Operator. If your group has collected all of your data, both partners are expected to work together on the analysis, but the role of Recorder will still alternate between partners.

- The Recorder notes what the group does during the session, records experimental conditions and preliminary results, and ensures that the data are securely saved. After each session (except the last one for a given experiment), the Recorder for that session will submit a Lab Notebook entry on Canvas on behalf of the group.

  The Recoder should also submit a Lab Note entry to describe sessions in which the group does analysis.

- The Operator interacts with the equipment, ensures all possible problems are addressed, suggests alternative approaches, and raises concerns with the suggested procedures, data quality, and preliminary results.

2.2 Text

There is no required text, but a text discussing error analysis, such as *An Introduction to Error Analysis* by J.R.Taylor, could be useful as a reference. This and similar texts on data analysis are listed at the end of this document, along with several general references.

Readings, equipment manuals, and references to the literature for each experiment will be provided in Canvas.

2.3 Assignments

2.3.1 Data Analysis Tutorial

At the beginning of the term, we will go through a series of in-class and homework exercises to become familiar with the steps used in data analysis.

The analysis steps and results should be detailed in an analysis note, and a summary of the results and conclusions should be written and submitted. Further details will be discussed in class.

2.3.2 Answer To Preliminary Questions

Before beginning each new experiment, students must individually submit answers to pre-lab questions listed in the lab manual. These will generally be due at 0800 on the first day of the experiment.

2.3.3 Lab Notes

The role of a laboratory notebook is to provide yourself and your collaborators with a complete record of your work, to keep all information in a central location, and to encourage critical thinking on your work [1]. Your entries must be sufficiently detailed so that you or someone else can conduct any procedure with only the notebook as a guide. Few students (and not that many researchers, for that matter) record sufficiently detailed and organized information. The raw data files produced by the data acquisition systems are stored locally on each computer. For each data file, be sure that you note the machine, directory, and name of the file in your notebook. In addition, your record of data files should include notes about the type of data collected and any special measurement conditions.

The "Lab Note" can either be a summary of what was done in the lab session, or it could be a direct copy of your notebook entries. Lab notes are due in Canvas at 2359 after each Lab session, with the exception of the last session of an experiment. The score for the Lab notes entry will apply to all members of the group. Feel free to check on each others work before submitting. The entries are evaluated on:

- Organization and Readability: How easy is following your work?
- Completeness: Do you Do you have enough detail to reconstruct your work? Do you describe the steps of your data-taking or analysis?
- Assessment of the data: Did you evaluate the quality of your data as you took them? Can you explain your data? Do you have a determination of the errors in your results?
- Plan for the next lab session.

[1] These ideas come from this pretty good blog entry. This article in *Science* also describes how to keep a good notebook.

2.3.4 Analysis Notes

As a group, prepare a short write-up that answers the objectives for the lab. You do not need to write a lot; answer the questions like homework questions. Your answers must be typed and submitted onto Canvas by the deadline. Once you submit your note, please make an appointment shortly with me. Both students need to attend the meeting. The final Recorder will lead the initial presentation of the analysis in about 8 minutes. Both students will answer questions for about 8 minutes. Grading:

- Your group gets full credit if the analysis is correct and the reporting is satisfactory (100%).
- If the reporting is unsatisfactory, your grade is decreased by 10% for each reporting awkwardness.
- If your analysis is correct but incomplete, your grade is decreased by 10% for each part missing. For example, you are asked to extract a coefficient for two energies; you extract it correctly but for only one energy.
- If none of the analysis is correct, your group scores a 55%.

Your group may improve the analysis and come again to present it. For each additional try, the satisfactory grade drops by 10%, and the non-satisfactory grade remains 55%. For example, if it takes two tries to get the analysis correct (and your reporting is satisfactory, and the analysis is complete), your final grade is 80%. The analysis note for an experiment needs to be presented before the deadline for submitting the Lab Report for that experiment.

2.3.5 Lab Reports & Peer Review

Each student will submit a technical report for each of the experiments. Although students will work in small groups to collect and analyse the data, the technical reports must be written by each individual student.

The style should follow the general guidelines of the "Physical Review Style and Notation Guide" and the "AIP Style Manual (Fourth Edition)", both of which are available online. Detailed report formatting and outline requirements are listed in Appendix A. I encourage you to learn to use the scientific word processor LATEX through Overleaf's web interface. But you are welcome to use any word processor that works for you.

The first three experiments will have a single draft of the lab report. For the fourth experiment, there will be both an initial draft due before the end of the semester, and a revised draft due during finals week. Each of these five reports will carry equal weight.

We will engage in peer reviews of the first three lab reports and the initial draft for the fourth experiment. The details of the peer review process will be discussed in class before we begin the first experiment.

2.3.6 Presentations

During our scheduled ”final exam” session, 12:20pm-2:20pm on Monday, 9 December, in Clippinger 105, each student will present a short (about 10 minute) talk on one of the experiments they have completed, or on a related topic. These will follow the general conventions of a contributed talk at an APS conference. The details of format and topics will be discussed around the middle of the semester.

2.3.7 Late work policy

APQs and Lab Notes received more than 24 hours after the deadline (without prior discussion) will not be graded and will receive a zero. Likewise, analysis notes, and reports received more than a week after the deadline will not be graded and will receive a zero. Remember that a university valid excuse abolishes these deadlines. In this case, contact me asap to discuss an alternative schedule. Furthermore, I will gladly extend deadlines to any reasonable and rare requests: ask me.

2.4 Grading Policy

The grading breakdown will be as follows:

Data analysis tutorial:	10%
Answers to preliminary questions:	10%
Lab notes:	10%
Analysis notes:	20%
Lab reports:	25%
Peer review:	5%
Final presentation:	10%
Participation:	10%

All assignments should be submitted by the end of the assigned final exam period: 2:20PM on Monday, 9 December. Any reports not submitted by that time, without prior agreement of the instructor, will receive a zero.

This course will not have a midterm exam. The presentations will take the place of the final exam. **The final presentations will be held from 12:20pm-2:20pm on Monday, 9 December, in Clippinger 105.**

3 Administrative policies

3.1 Attendance

Attendance during scheduled laboratory and class times is critical in completing the required work. Students should arrive punctually for all class sessions, and stay for the full period. If you will need to miss a class session for reasons other than illness, please contact the instructor before the class

session. Chronic tardiness, leaving class early, or absence will reduce your participation grade. If the student has a University excuse, accommodations will be made; please discuss this with the instructor.

If you are ill, stay home. If your symptoms persist, contact Health Services. Contact the instructor by phone or email as soon as you are able to do so, to determine how best to keep up with the class.

In addition to the attendance policies already listed, you may be absent for up to three (3) days each academic semester, without penalty, to take time off for reasons of faith or religious or spiritual belief system or to participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization. **You are required to notify the instructor in writing of specific dates requested for alternative accommodations no later than fourteen (14) days after the first day of instruction. These requests will remain confidential.** For more information about this policy, you may contact the Director and Title IX Coordinator, Equity and Civil Rights Compliance, Lindley Hall 006, 740-593-9140, Equity@ohio.edu.

3.2 Accessibility

Any student who feels they may need an accommodation based on the impact of a disability should contact me privately to discuss your specific needs and provide written documentation from Accessibility Services. If you are not yet registered as a student with a disability, please contact Accessibility Services at 740-593-2620 or visit the office in Alden Library, Suite 230.

3.3 Use of technologies like equation solvers, ChatGPT, or AI

Equation solvers, ChatGPT, or other AI technologies can be useful tools in the process of understanding the mathematical and physical processes we are discussing in the course, but are not a substitute for developing your own understanding. You are welcome to use these or related technologies in completing assignments, but you must indicate how these tools informed your process and the final product.

Using these tools to generate content without proper attribution qualifies as academic dishonesty.

3.4 Academic honesty

Academic misconduct is a serious offense against the Student Code of Conduct, and can lead to sanctions up to and including expulsion from the university. The Student Code of Conduct (http://www.ohio.edu/communitystandards/academic/students.cfm) defines academic misconduct to include: plagiarism; cheating; submitting the same academic work more than once without consent of the instructor; fabrication (e.g., use of invented information or falsification of research or other findings); and misrepresenting facts in an attempt to gain an academic advantage.

Plagiarism can be the most subtle of the offenses, and can happen inadvertently, through improper citation. Unintentional plagiarism occurs when material is copied or copied with minor changes without properly indicating it as another's work. If you are unsure whether a citation is required, you should include a citation.

A student caught in a flagrant example of academic misconduct may be given a zero on the project or the course, and the case may be referred to the Office of Community Standards and Student Responsibility. The policies, procedures, and sanctions which follow are detailed in the Student Code of Conduct. In addition, all cases of flagrant misconduct or suspected misconduct will be reported in writing to the Chairman of the Department of Physics and Astronomy.

3.5 Protection of intellectual property

The lectures, classroom activities and all materials associated with this class and developed by the instructor are copyrighted in the name of Paul King on this date, 26 August 2024.

A PHYS6751 Report Formatting Requirements

Your report must conform to the following requirements. The document will be rejected and not graded if these requirements are not respected. If you submit your report too close to the deadline (+ late policy allowance) and it is rejected because it doesn't respect the format requirement, you might get a 0 for this assignment.

- Your report needs to follow exactly the table of content listed below.
- The experiment title, the author name, and the name of their partners need to appear at the top of the first page in lines containing this information only.
- Your report needs to be between 2250 and 2750 words, about 5 pages of text. Each section should be between 400 and 800 words. Show your word count.
- The tables and figures should have a label and caption, and be referred to in the text.
- You need to number the lines of your report. In Latex, use the linenumbers command. In Word, on the Layout tab, in the Page Setup group, click Line Numbers.
- The document should be written in single spaced format, with no more than six lines of text within a vertical space of one inch. Margins, in all directions, must be at least an inch.
- Use one of the following typefaces (i) Arial, Courrier New, Palatino Linotype at a font size between 10 and 12 points; (ii) Times New Roman at a font size of 11 or 12 points or (iii) Computer Modern family of fonts at a font size of 11 or 12 points. This document is written in the Lucida Computer Modern font at 12-point size.

Required table of contents for the reports in PHYS 6751:

Abstract
I. Background Introduction
    I.A. Motivation
    I.B. Summary of experiment
II. Theory
III. Experimental Methods
    III.A. Description of apparatus
    III.B. Description of experimental procedure
    III.C. Description in plain English of the observed phenomena
    III.D. Description of sources of errors and how you minimize them
IV. Analysis and Results
    IV.A. Description of the data set
    IV.B. Method of Analysis
    IV.C. Presentation of results
    IV.D. Discussion of results
V. Conclusions
    V.A. Summary of results
    V.B. Pertinence of results to the questions raised in the introduction
    V.C. Further questions raised by this experiment. Suggestion for future work
References

B Possible References

B.1 Nuclear Data

1. National Nuclear Data Center, information extracted from the NuDat database, `https://www.nndc.bnl.gov/nudat/`

2. R.B. Firestone, V.S. Shirley, editors, *Table of Isotopes,* 8th ed. (John Wiley & Sons, New York, 1996).

3. E. Browne, R.B. Firestone, *Table of Radioactive Isotopes* (John Wiley & Sons, New York, 1986).

4. National Nuclear Data Center, `http://www.nndc.bnl.gov/`

B.2 Detectors and Electronics

1. W.R. Leo, *Techniques for Nuclear and Particle Physics Experiments: A How To Approach,* (Springer, Berlin, 1987).

2. G.F. Knoll, *Radiation Detection and Measurement,* 2nd ed. (John Wiley & Sons, New York, 1969);
   G.F. Knoll, *Radiation Detection and Measurement,* 3rd ed. (John Wiley & Sons, New York, 2000).

3. D. Green, *The Physics of Particle Detectors,* (Cambridge University Press, Cambridge, UK, 2000).

B.3 Solid State and Quantum Physics

1. C. Kittel, *Introduction to Solid State Physics,* 8th ed. (John Wiley & Sons, New York, 2004).

2. R. Eisberg and R. Resnick, *Quantum Physics of Atoms, Molecules, Solids, Nuclei, and Particles,* 2nd ed. (John Wiley & Sons, New York, 1985).

3. K. Krane, *Modern Physics* 2nd ed. (John Wiley & Sons, New York, 1995);
   K. Krane, *Modern Physics* 3rd ed. (John Wiley & Sons, New York, 2012)

B.4 Data Analysis & Error Analysis

1. P.R. Bevington, *Data Reduction and Error Analysis for the Physical Sciences,* (McGraw-Hill, New York, 1969);
   P.R. Bevington and D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences,* 3rd ed. (McGraw-Hill, New York, 2002).

2. J.R. Taylor, *An Introduction to Error Analysis,* 2nd ed. (University Science Books, Sausalito, CA, 1997).

3. L. Lyons, *Statistics for Nuclear and Particle Physicists,* (Cambridge University Press, Cambridge, UK, 1986).

4. L. Kirkup, *Data Analysis with Excel,* (Cambridge University Press, New York, 2002).

5. R. de Levie, *Advanced Excel for scientific data analysis,* (Oxford University Press, New York, 2004).

B.5 Citations, Writing, and Editing

1. "APS Journals Style Guide for Authors", `https://res.cloudinary.com/apsphysics/image/upload/v1715884920/aps-journals-style-guide_tnoyln.pdf`

2. "AIP Publishing Author Instructions", `https://publishing.aip.org/resources/researchers/author-instructions/`

3. W. Strunk Jr. and E.B. White, *The Elements of Style,* 3rd ed. (Macmillan, New York, 1979).

4. J. Gibaldi, *MLA Handbook for Writers of Research Papers,* 6th ed. (MLA, New York, 2003).

5. R.A. Day and B. Gastel, *How to Write and Publish a Scientific Paper,* 6th ed. (Greenwood Press, Westport, CT, 2006).

6. C. Lipson, *Doing Honest Work in College,* (University of Chicago Press, Chicago, IL, 2004).